ISA² Programme & EIF



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G DIGIT European Commission

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Brussels, Belgium
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Connecting public administrations, businesses and citizens

ISA² Programme

EIF: European Interoperability Framework

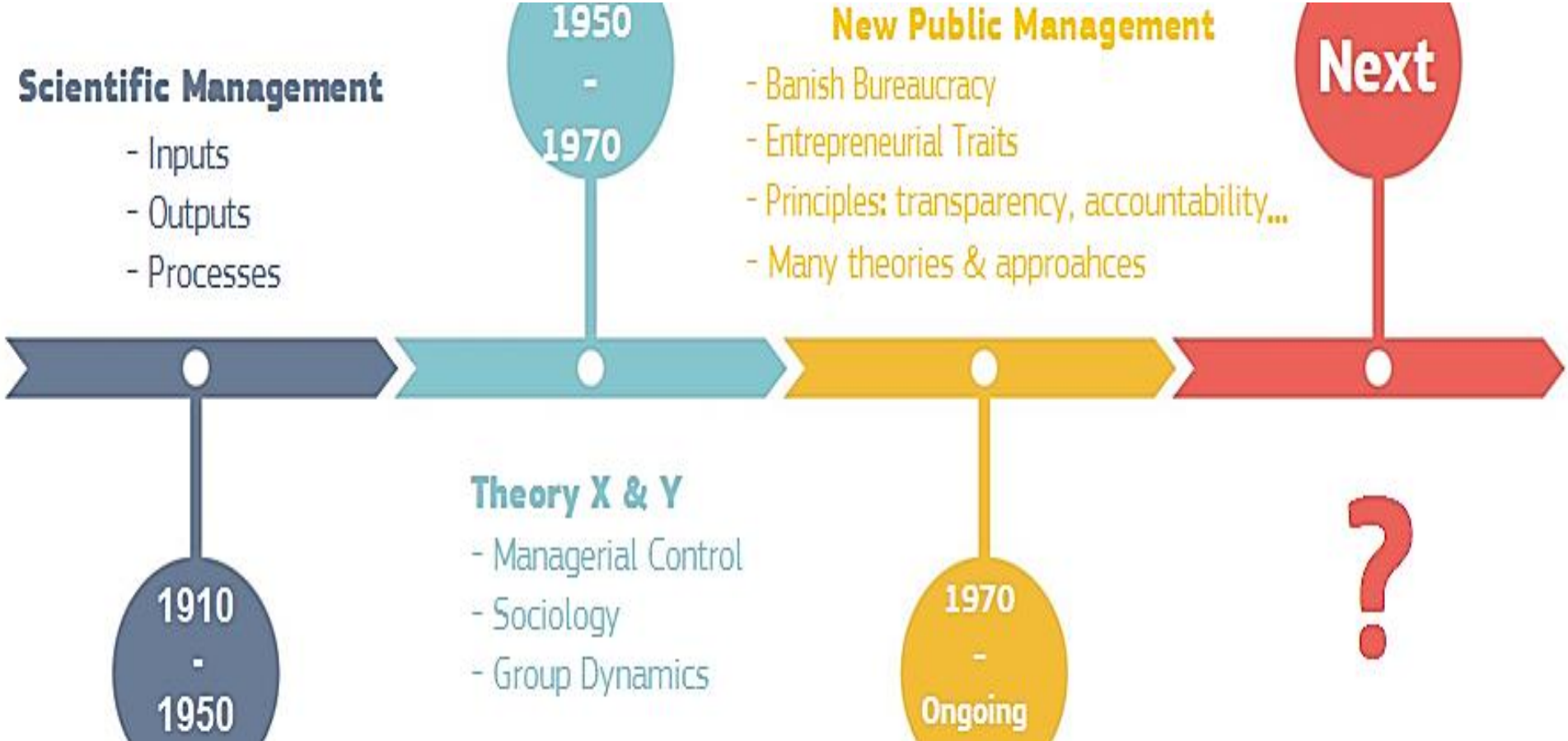
EIRA: European Interoperability Reference Architecture

Connecting public administrations, businesses and citizens



DIGITAL is everywhere.
It should be used as a key enabler in any Public Service

THE PUBLIC SECTOR HAS BEEN TRANSFORMING SINCE THE BEGINNING





By 2025, everyone on Earth is expected to have internet access.



ISA²

Programme

INTEROPERABILITY SOLUTIONS FOR PUBLIC ADMINISTRATIONS, BUSINESSES AND CITIZENS

European Commission

ISA² PROGRAMME

More than 30 years digitising public administrations across the EU



1985-1992	1995	1999	2005	2010	2016	2020
CADDIA PROGRAMME	**IDA Programme** Interchange of Data between Administrations	**IDAII Programme** Follow-on programme of IDA	**IDABC Programme** Interchange of Data between Administrations, Businesses and Citizens	**ISA Programme** Interoperability Solutions for Public Administrations	**ISA² Programme** Interoperability Solutions for Public Administrations, Businesses and Citizens	

European Commission

ISA² Programme solutions

Check out our interoperability and open source solutions to cut red tape. More to come!

Key & generic interoperability enablers

- **eTrustEx** — Secure document exchange platform
- **e-Docs** — ArchiMate diagrams visualising e-Document lifecycle
- **CPSV-AP** — Data model to describe public services & associated life & business events
- **TESTA** — Data communication network service

Semantic interoperability

- **DCAT-AP** — Metadata standard
- **Core Vocabularies** — Share your own semantic assets
- **ADMS** — Government metadata management
- **VocBench3** — Multilingual platform for collaborative thesaurus management

Decision making & legislation

- **LEOS** — Facilitate the editing of legislative texts
- **OCS for ECIs** — Gather statements of support online

EU policies -Supporting instruments

- **ePrior** — e-Procurement system for Public Administrations
- **ESPD** — Facilitate participation in public procurement across the EU

e-Procurement/ e-Invoicing

- **e-Certis** — Map certificates necessary in public procurement
- **Re3gistry** — Manage & share reference codes

Geospatial solutions

- **CAMSS** — Common assessment method for standards & specifications
- **Joinup** — Platform for sharing & reuse of IT solutions for public administrations

Supporting instruments for public administrations

- **CIRCABC** — Communication & Info Resource Centre for Administrations, Businesses & Citizens
- **NIFO** — National Interoperability Framework Observatory
- **EIRA & EICart** — EU IOP Reference Architecture & EU Interoperability Cartography
- **PM²** — Open Project Management Methodology
- **EUSurvey** — Create, manage and analyse online surveys & public consultations
- **EUPL** — Open source licence
- **IMM** — Interoperability Maturity Model
- **IOP TestBed** — Interoperability & conformance testing platform

You click, we link!

Learn more:

ISA²



EIF

EUROPEAN INTEROPERABILITY FRAMEWORK

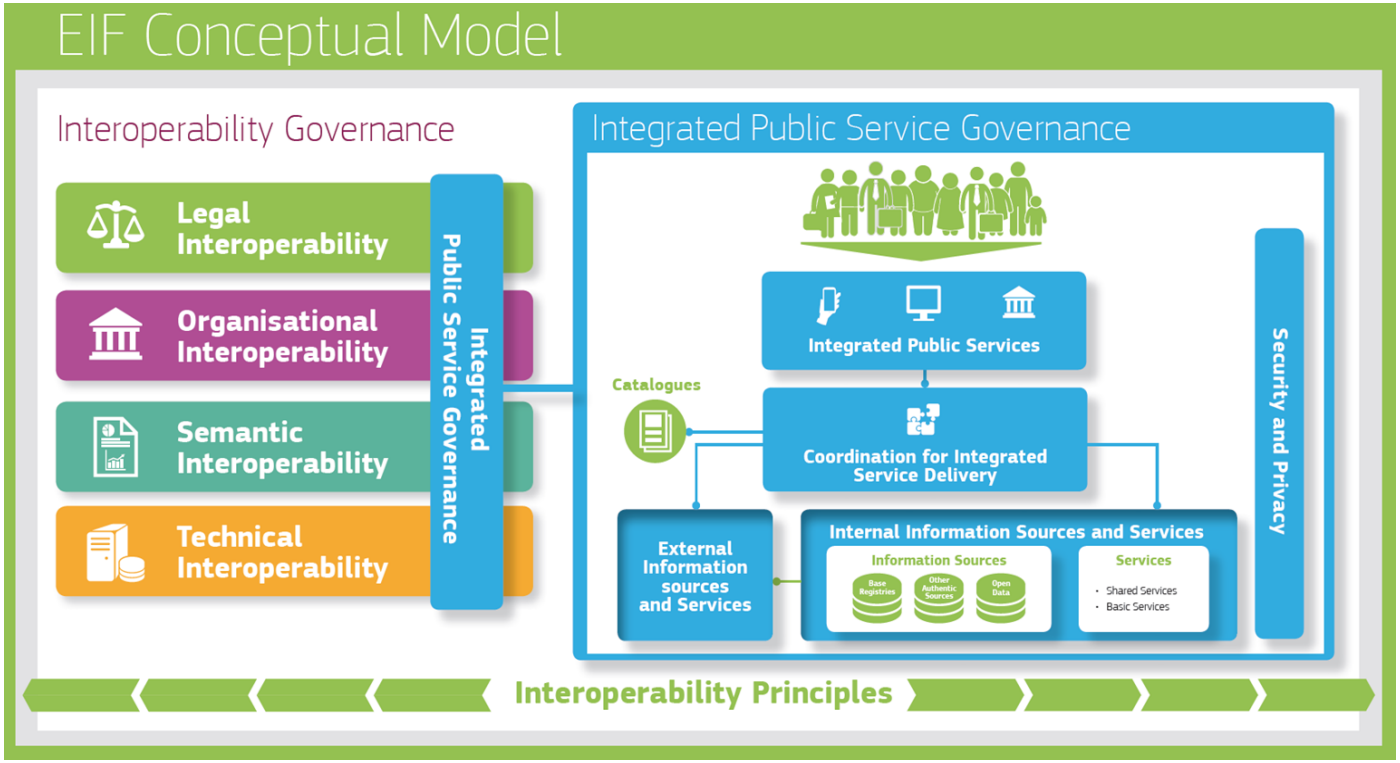
Connecting public administrations, businesses and citizens

EIF (European Interoperability Framework)

European Commission

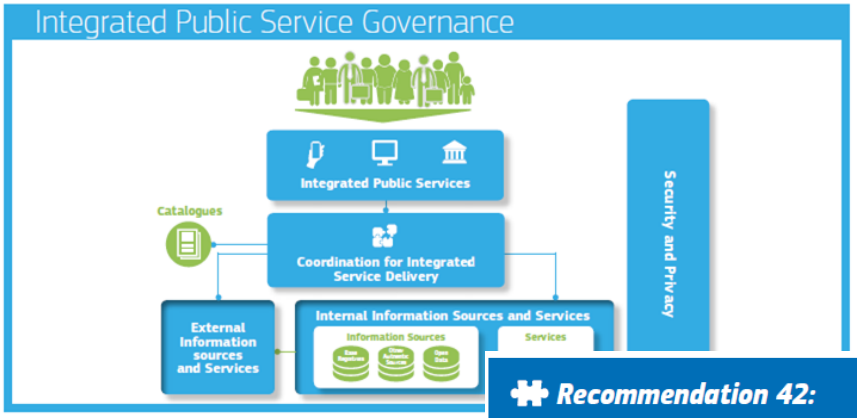
EIF

- 1 conceptual model
- 4 levels of interoperability
- 12 underlying principles
- 47 recommendations



ISA²

EIF from abstract to practical



Use specs such as DCAT-AP (or extensions i.e. GeoDCAT-AP), CPSV, ADMS to describe open data, public services and interoperable solutions

Recommendation 42:

Publish open data in machine-readable, non-proprietary formats. Ensure that open data is accompanied by high quality, machine-readable metadata in non-proprietary formats, including a description of their content, the way data is collected and its level of quality and the licence terms under which it is made available. The use of common vocabularies for expressing metadata is recommended.

Recommendation 44:

Put in place catalogues of public services, public data, and interoperability solutions and use common models for describing them.

New EIF benefits

European Commission

Public administrations

- More data and of good quality for better decision making
- Sharing and reuse to avoid vendor lock-in
- More credibility and trust

Citizens

- Easy to identify and use the needed public service
- Submission of data once-only
- Tracking progress and giving feedback
- More participation

Businesses

- Easy access to public services
- Easy to fulfil reporting obligations
- New opportunities by using open public data

Cost and time savings

ISA²

European Commission

Interoperability principles

1: Subsidiarity and proportionality
1 recommendation

2: Openness
3 recommendations

3: Transparency
1 recommendation

4: Reusability
2 recommendations

5: Technological neutrality and data portability
2 recommendations

6: User-centricity
4 recommendations

7: Inclusion and accessibility
1 recommendation

8: Security and privacy
1 recommendation

9: Multilingualism
1 recommendation

10: Administrative simplification
1 recommendation

11: Preservation of information
1 recommendation

12: Assessment of Effectiveness and Efficiency
1 recommendation

Reuse and EIF

Reuse & share at the core

Underlying principle 4: reusability

There are some key challenges that limit the sharing and reuse of IT solutions, at technical, organisational, legal and communication levels. **The ISA² sharing and reuse framework for IT solutions**[12] **provides recommendations for public administrations** to help them overcome these challenges and share/ reuse common IT solutions. Reuse and sharing can be effectively supported by collaborative platforms.[13]

Recommendation 6:
Reuse and share solutions, and cooperate in the development of joint solutions when implementing European public services.

Recommendation 7:
Reuse and share information and data when implementing European public services, unless certain privacy or confidentiality restrictions apply.

Sharing and Reuse of IT solutions Framework

- New to public administrations on how they can create reusable IT solutions and how they can benefit from reusing already existing ones
- Should be used at all development stages of IT solutions & all levels of public administrations



Identifies 4 main types of barriers

10 key recommendations

13 recommended measures for central bodies

16 supporting instruments

joinup

Share and reuse Interoperability solutions for public administrations, businesses and citizens

Start typing to search...

92
Collections
Collaborate around thematics

2706
Solutions
Interoperability solutions ready to use

16204
Events, discussions, news ...
To keep you up to date

Joinup is a collaborative platform created by the European Commission and funded by the European Union via the Interoperability solutions for public administrations, businesses and citizens (ISA²) Programme. It offers several services that aim to help e-Government professionals share their experience with each other. We also hope to support them to find, choose, re-use, develop and implement interoperability solutions.

PM² presentation
Joinup Moderator
13/10/2017

I will have the honour to present PM² and the Open PM² initiative during the "Rencontres du Management de Projet"

news

EIRA
EUROPEAN INTEROPERABILITY REFERENCE ARCHITECTURE

EIRA

0 solution

Brand new Joinup
Joinup Moderator
04/10/2017

Welcome to the brand new Joinup! Joinup, the European Commission's collaboration platform for digital government

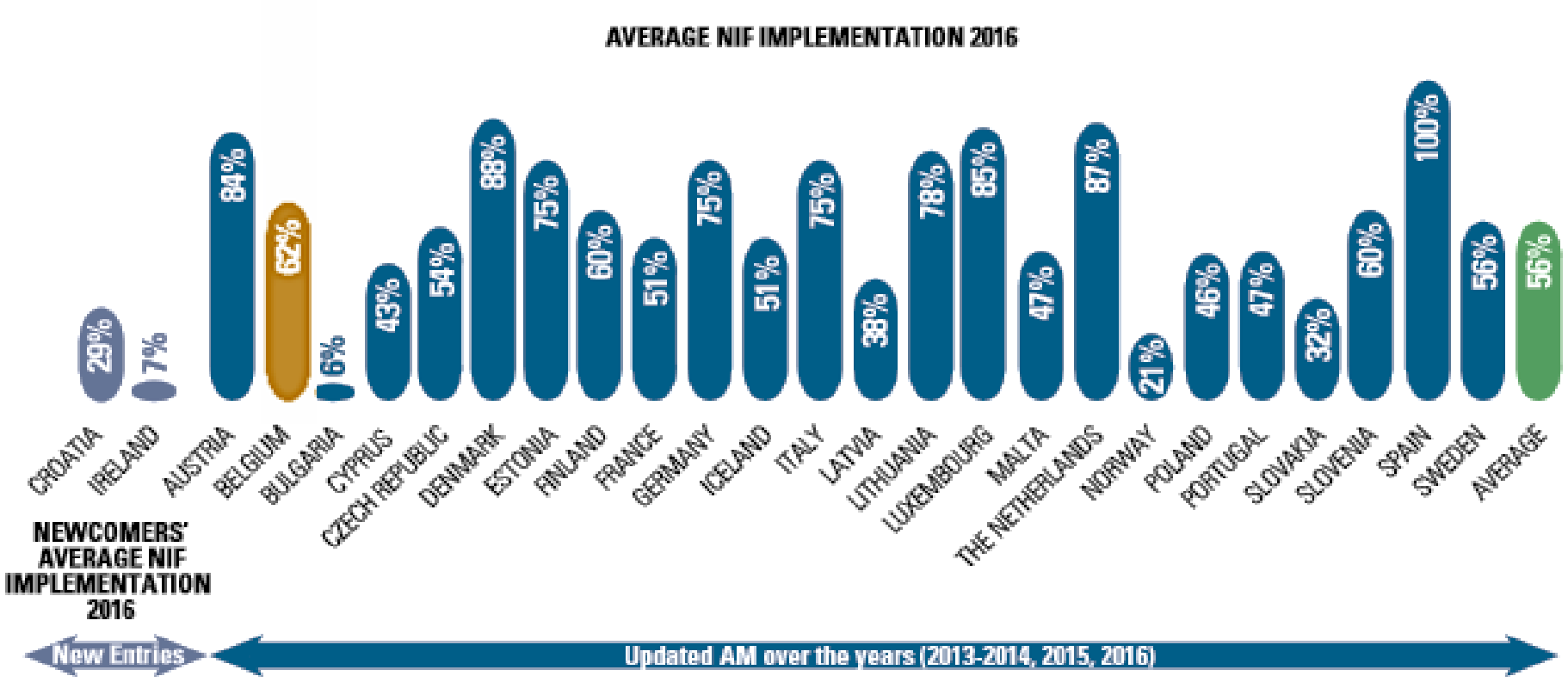
new joinup launch

news



EUPL

108 collection


AVERAGE NIF IMPLEMENTATION 2016
29%
7%
84%
62%
6%
43%
54%
88%
75%
60%
51%
75%
51%
75%
38%
78%
85%
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87%
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46%
47%
32%
60%
100%
56%
56%
CROATIA
IRELAND
AUSTRIA
BELGIUM
BULGARIA
CYPRUS
CZECH REPUBLIC
DENMARK
ESTONIA
FINLAND
FRANCE
GERMANY
ICELAND
ITALY
LATVIA
LITHUANIA
LUXEMBOURG
MALTA
THE NETHERLANDS
NORWAY
POLAND
PORTUGAL
SLOVAKIA
SLOVENIA
SPAIN
SWEDEN
AVERAGE
NEWCOMERS' AVERAGE NIF IMPLEMENTATION 2016
New Entries
Updated AM over the years (2013-2014, 2015, 2016)

European Commission

European Interoperability Framework

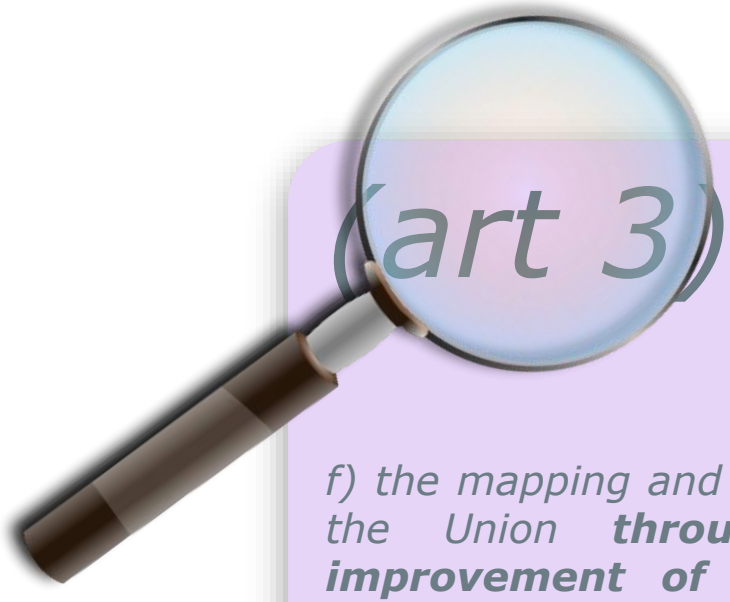
Roadmap for the digitalisation of Public Administrations



ISA²


European Commission
EIRA
European Interoperability Reference Architecture
Building efficient digital public services

EIRA. Why EIRA? (*)



(art 3) "…. The ISA²programme shall support and promote:

f) the mapping and analysis of the overall interoperability landscape in the Union ***through the establishment, maintenance and improvement of the EIRA and the EIC as instruments to facilitate the re-use of existing interoperability solutions and to identify the areas where such solutions are still lacking****;*

(g) ***the maintenance, updating, promotion and monitoring of the implementation of the EIS, the EIF and the EIRA***"

(*) DECISION (EU) 2015/2240 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 25 November 2015 http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32015D2240

EIRA benefits





Designing

Accelerate the design of systems that support the delivery of interoperable digital public services (across borders and sectors).



Assessing

Provide a reference model for comparing existing architectures in different policy domains and thematic areas, to identify focal points for convergence and reuse.



Communicating and Sharing

Help documenting the most salient interoperability elements of complex systems and facilitate the sharing of reusable solutions.



Discovering and Reusing

Ease the discovery and reuse of interoperability solutions through the TES Cartography using the CarTool.



EIF - EIRA alignment

European Commission

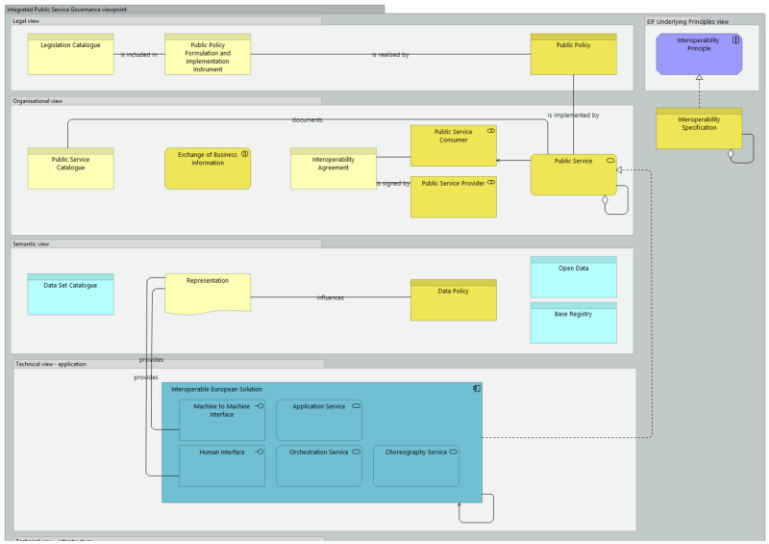
EIF

Framework and principles



EIRA

Implementation guidelines and specifications

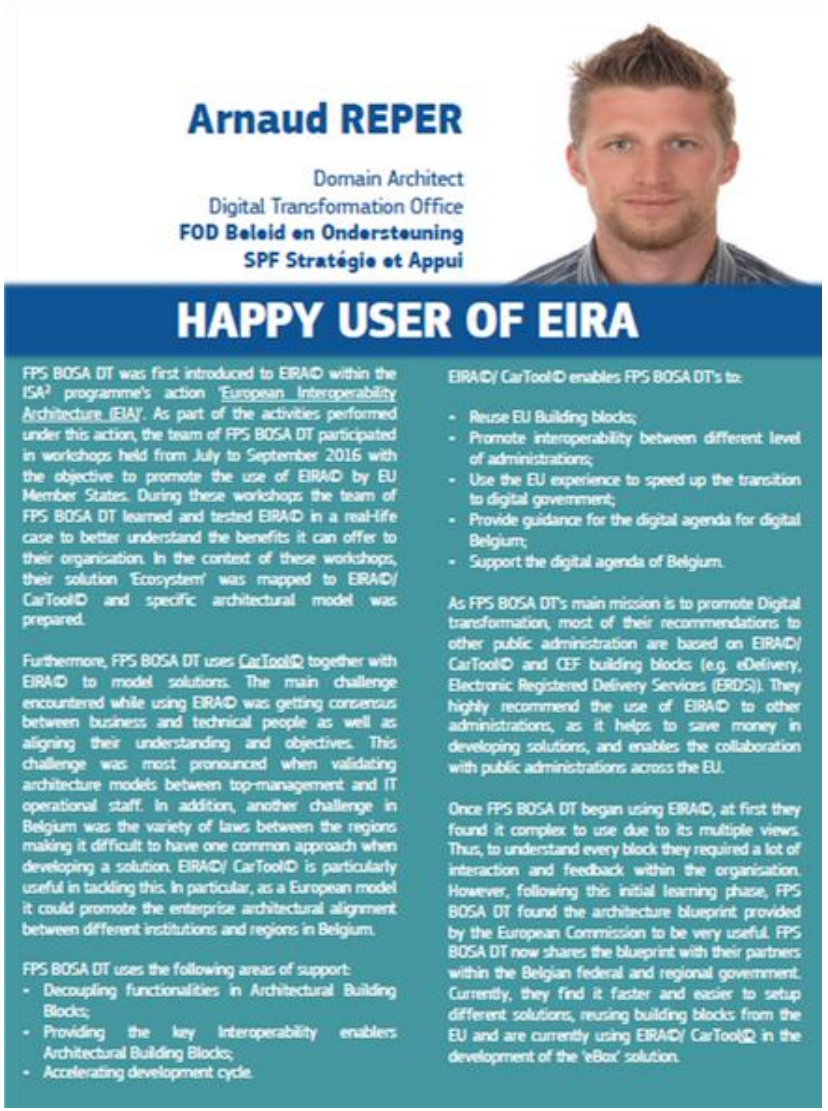


ISA²

Always with they eye on the users

"EIRA© is particularly useful in tackling this. In particular, as a European model it could promote the enterprise architectural alignment between different institutions and regions in Belgium".

"As FPS BOSA DT's main mission is to promote Digital transformation, most of their recommendations to other public administration are based on EIRA© and CEF building blocks -e.g. eDelivery, Electronic Registered Delivery Services



Arnaud REPER

Domain Architect
Digital Transformation Office
FOD Beleid en Ondersteuning
SPF Stratégie et Appui

HAPPY USER OF EIRA

FPS BOSA DT was first introduced to EIRA© within the ISA² programme's action 'European Interoperability Architecture (EIA)'. As part of the activities performed under this action, the team of FPS BOSA DT participated in workshops held from July to September 2016 with the objective to promote the use of EIRA© by EU Member States. During these workshops the team of FPS BOSA DT learned and tested EIRA© in a real-life case to better understand the benefits it can offer to their organisation. In the context of these workshops, their solution 'Ecosystem' was mapped to EIRA©/ CarTool© and specific architectural model was prepared.

Furthermore, FPS BOSA DT uses CarTool© together with EIRA© to model solutions. The main challenge encountered while using EIRA© was getting consensus between business and technical people as well as aligning their understanding and objectives. This challenge was most pronounced when validating architecture models between top-management and IT operational staff. In addition, another challenge in Belgium was the variety of laws between the regions making it difficult to have one common approach when developing a solution. EIRA©/ CarTool© is particularly useful in tackling this. In particular, as a European model it could promote the enterprise architectural alignment between different institutions and regions in Belgium.

FPS BOSA DT uses the following areas of support:

- Decoupling functionalities in Architectural Building Blocks;
- Providing the key Interoperability enablers Architectural Building Blocks;
- Accelerating development cycle.

EIRA©/ CarTool© enables FPS BOSA DT's to:

- Reuse EU Building blocks;
- Promote interoperability between different level of administrations;
- Use the EU experience to speed up the transition to digital government;
- Provide guidance for the digital agenda for digital Belgium;
- Support the digital agenda of Belgium.

As FPS BOSA DT's main mission is to promote Digital transformation, most of their recommendations to other public administration are based on EIRA©/ CarTool© and CEF building blocks (e.g. eDelivery, Electronic Registered Delivery Services (ERDS)). They highly recommend the use of EIRA© to other administrations, as it helps to save money in developing solutions, and enables the collaboration with public administrations across the EU.

Once FPS BOSA DT began using EIRA©, at first they found it complex to use due to its multiple views. Thus, to understand every block they required a lot of interaction and feedback within the organisation. However, following this initial learning phase, FPS BOSA DT found the architecture blueprint provided by the European Commission to be very useful. FPS BOSA DT now shares the blueprint with their partners within the Belgian federal and regional government. Currently, they find it faster and easier to setup different solutions, reusing building blocks from the EU and are currently using EIRA©/ CarTool© in the development of the 'eBox' solution.



SEMIC 2018

Semantic Interoperability Conference

Linked Digital Public Administrations

14 June 2018 - Sofia, Bulgaria



#SEMIC201
@EU_isa2

Initiative supported by the ISA² Programme



Questions?

ISA² programme

You click, we link

Run by the ISA Unit at DIGIT (European Commission) with 131€M budget, the ISA² programme provides public administrations, businesses and citizens with specifications and standards, software and services to reduce administrative burdens.

@EU_ISA2

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ec.europa.eu/isa2